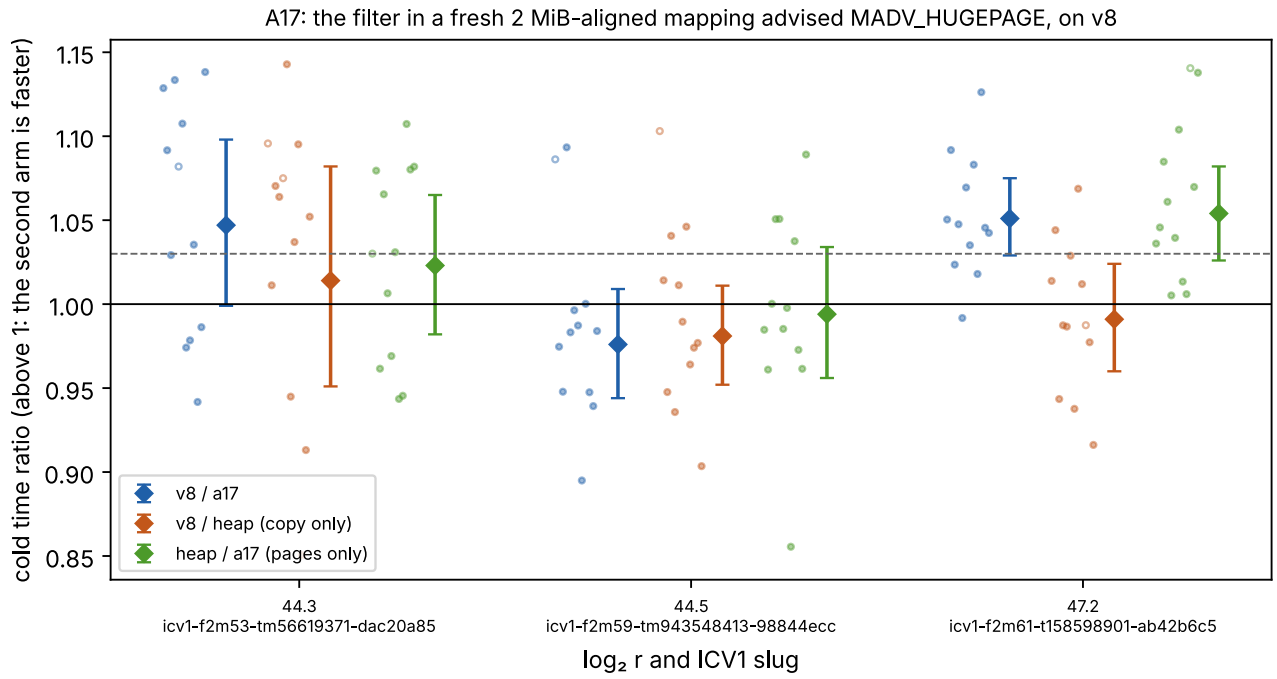
A17: the presence filter on huge pages from its first touch

Exploration, no round. v8 (`ic-r12-v7-23138d20`) against v8 with the filter copied into a fresh 2 MiB-aligned anonymous mapping, advised `MADV_HUGEPAGE` before it is written (`ic-a17-v8-fb4284d8`, `filter-huge-pages.patch`). A third arm, `heap`, is the same binary with `KIC_FILTER_HUGE=0` (the copy alone).

The timed runs: 2026-10-17 05:08–05:35 UTC; 108 isolated processes, all completed. Four were contended and are excluded. Every logarithm is the same in all three arms.

The stage probes: R04's probes on both arms, interleaved; 48 processes.

Outcome: a gain at 2^47.2 only, so no round. A round needs every target size above 1.03.



explorations/A17-filter-huge-pages-20261017/pairs-*.tsv; 108 isolated processes; open: a contended run; dashed: 1.03

Figure 1: Cold-time ratios per pair (open: a contended run) and the clean pairs' geometric mean with its 95% t interval. Dashed: 1.03.

curve	$\log_2 r$	v8 / a17 [95%]	v8 / heap	heap / a17	filter stage, v8 → a17 (ns)
`icv1-f2m53-tm56619371-dac20a85`	44.3	1.047 [0.999, 1.098]	1.014	1.023	4.75 → 4.78
`icv1-f2m59-tm943548413-98844ecc`	44.5	0.976 [0.944, 1.009]	0.981	0.994	4.92 → 4.56
`icv1-f2m61-t158598901-ab42b6c5`	47.2	1.051 [1.029, 1.075]	0.991	1.054 [1.026, 1.082]	6.43 → 5.32

Reading:

- **The copy costs nothing measurable,** and the gain at 2^47.2 is the pages'.
- **The gain follows the filter's size.** The 8 MiB filter spans 2,048 base pages, as many as this core's second-level TLB holds. The 4 MiB filters span 1,024 base pages and show no consistent change.
- **This is a hypothesis.** The VM exposes no performance counters, so TLB misses were not counted.
- **Huge pages were confirmed** through `AnonHugePages` on unisolated diagnostic runs: 4,096 kB and 8,192 kB.

Scope: one host, one core, three sizes, two rows a size. THP is `madvise` here.